

CV 5101
Modeling, Uncertainty, and
Data for Engineers
(July – Nov 2026)

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L7
Sep 16, 2026

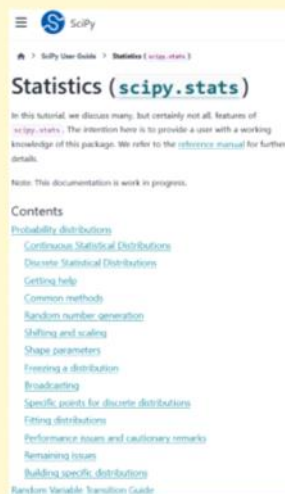
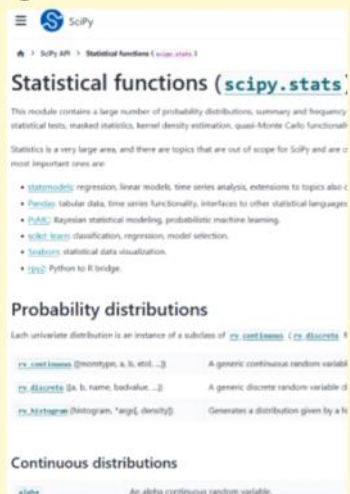
Flow

- Revise
 - Variance, covariance, **correlation**
 - Gaussian
 - reading z-table
 - Expectation
- Parametric distributions
 - Gaussian, Uniform, exponential, lognormal, Gumbel
- Propagation laws

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Requested practice?

- Practice **scipy.stats** for the following:
 - Using distributions → Shifting, scaling
 - Generating random numbers
 - Fitting distributions



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Module Overview

B.1

Uncertainty & Estimation

B.1.1

Probabilistic basics and RVs

- Probability laws
- Random variables
- Discrete, continuous RVs
- PMF/PDF/CDF
- Expectation
- Variance/cov./corr.

Q1C5, Funda6-7-8

B.1.2

Core distributions

- Gaussian
- Uniform
- Exponential
- Lognormal
- Symmetric/skewness
- *Tails*: thick/light

Q1C5, Funda9-10

B.1.3

Propagation laws & least-sq.

- Uncertainty propagation
- Least-square errors
- Parameter estimation

Q1C2-C3, Funda9-10

Revise

- Two structural engineers, Alice and Bob, independently measure the wind speed in m/s at the top of each tower.

Alice's reading $X \sim \mathcal{N}(\mu = 40, \sigma = 5)$

Bob's reading $Y \sim \mathcal{N}(\mu = 42, \sigma = 6)$

Their readings are correlated with $\rho = 0.8$

1. If Alice and Bob average their measurements, $W = (X + Y)/2$, what are $E[W]$ and $\text{Var}[W]$?
2. What's the covariance of X and Y ?
3. If design wind speed is 55 m/s, what's the prob. that the average exceeds 55 m/s?

$$\begin{aligned} \mu_W &= E[W] = E\left[\frac{X+Y}{2}\right] \\ &= \frac{1}{2} E[X] + \frac{1}{2} E[Y] = 41 \end{aligned}$$



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Revise

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Check-in with teach-book

<https://mude.citg.tudelft.nl/book/2024/>

- Q1 Topics (Chapters 5, 6, 2, and 3)
 - **Q1C5 Univariate continuous distribution**
 - **Q1C5.1** PDF/CDF
 - **Q1C5.2** Empirical Distributions
 - **Q1C5.3** Parametric Distributions
 - Q1C5.4 Fitting a Distribution
 - Q1C6 Multivariate Distributions (briefly)
 - **Q1C2** Propagation of Uncertainty
 - Q1C3 Observation Theory: least-sq., Hyp. Test, Conf. Intervals
- Q2 Topics (Chapter 7 and 8)
 - Q2C7 Extreme value theory: GEV, return period, POT
 - Q2C8 Risk and decision making (CBA)
- Fundamental Concepts
 - **Chapter 6, 7, 8, 9. Probability basics, rv, z- and t-tables**
- Programming
 - Fundamental Concepts → Chapter 10

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L8

Sep 17, 2026

Recap

- **Probability and RV:** Probability Rules, Total probability Rule, Bayes' Rule
- **Core distributions:**
 - Gaussian/Normal, "Transformation"
 - PDF, CDF, mean, stdev, skewness, kurtosis
 - Uniform, Lognormal, Gumbel, Exponential
- **Transformation:** X 's PDF $\xrightarrow{Y=R(X)} \text{Jacobian} \rightarrow Y$'s PDF
- **Uncertainty classification:**
 - Aleatory: Inherent, Irreducible
 - Epistemic: Lack of knowledge/model
 - *helpful dichotomy; Lines are a bit bit blurry.*

Four Categories of Prediction

1. **Analytical**: (PPP, + PSB)
you know the physics and can obtain exact solution.
e.g., Newton's equation for falling ball; Jacobian's method
2. **Approximate**: (TODAY)
you know the physics but exact solution is either impractical/unavailable
you approximate the problem.
e.g., FEM, Taylor's Series, Linearization.
3. **Simulation**: (ASSTJ)
you know the physics but direct evaluation with full uncertainty is too difficult
e.g., Monte Carlo (Gold standard)
4. **Data-driven Prediction**: (SH)
Learn from the relationship in data. Pattern recognition.
e.g., Neural Networks, LLMs

10 variables, each with 10 possible values $\Rightarrow$ 10 Billin
Interesting to: "Combine Physics, Approximation,
Simulation, & Data-Driven Learning"

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B.1.3 Propagation laws & least-squares (Q1C2)

Uncertainty classification

Sources of Uncertainty

- **Inherent uncertainty**: concrete strength; irreducible
- **Statistical uncertainty**: due to the lack of data; reducible
- **Characteristics of system**: material properties, dimensions
- **Demand** placed on the system
- **Mathematical models** used to analyze structure's behavior
- **Measurements uncertainty**: while assessing structure's health
- **Probabilistic models** used to describe uncertain quantities
- **Human error**

Categories of Uncertainty

In classical statistics,

- Aleatory *Inherent, Irreducible*

- ^{epistemology}
• Epistemic
 - Reducible
 - Due to lack of knowledge

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Categories of Uncertainty

- Wind speed on the surface of a building:
 - Aleatory or epistemic?
- Case 1: When there is no predictive wind model and only *long-term statistical data is used?*
- Case 2: When wind fields are studied using *precise computational fluid dynamics?*

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Propagation Laws

Mean and Variance propagation laws

- A random variable, $X \sim \mathcal{N}(40, 5)$ *std dev*

If we define, another random variable, $Y = 3X + 10$, what are its mean and standard deviation (std dev) and type.

$$\begin{aligned} E[Y] &= E[3X + 10] \leftarrow \text{linear operator} \\ &= E[3X] + E[10] \\ &= 3E[X] + 10 = 3\mu_X + 10 = 130 \end{aligned}$$

Recall:

$$\begin{aligned} \text{Var}[Y] &= E[(Y - \mu_Y)^2] \\ &= E[\{(3X + 10) - (3\mu_X + 10)\}^2] \\ &= E[9(X - \mu_X)^2] \leftarrow \text{linear operator} \\ &= 9 E[(X - \mu_X)^2] = 9 \text{Var}[X] \end{aligned}$$

Mean and Variance propagation laws

- More generally, if we have $X_1, X_2, \dots, X_n$ with means $\mu_1, \mu_2, \dots, \mu_n$ and stdevs $\sigma_1, \sigma_2, \dots, \sigma_n$.
- For a $Y = q(X) = q(X_1, X_2, \dots, X_n)$, what are its expected value and variance?

$$Y = a_1 X_1 + a_2 X_2 + c$$

$$E[Y] = a_1 \mu_1 + a_2 \mu_2 + c$$

$$\text{Var}[Y] = E[(Y - \mu_Y)^2] = E[\{(a_1 X_1 + a_2 X_2 + c) - (a_1 \mu_1 + a_2 \mu_2 + c)\}^2]$$

$$= E[\{a_1(X_1 - \mu_1) + a_2(X_2 - \mu_2)\}^2]$$

$$= E[a_1^2(X_1 - \mu_1)^2 + a_2^2(X_2 - \mu_2)^2 + 2a_1a_2(X_1 - \mu_1)(X_2 - \mu_2)] \quad a_1=1, a_2=1$$

$$= a_1^2 E[(X_1 - \mu_1)^2] + a_2^2 E[(X_2 - \mu_2)^2] + 2a_1a_2 \text{COV}(X_1, X_2)$$

$$= a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + 2a_1a_2 \rho \sigma_1 \sigma_2$$

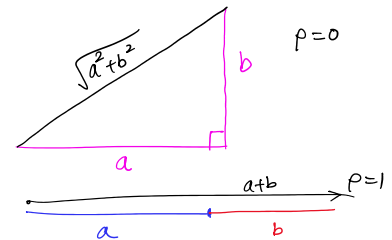
$$= a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + 2a_1a_2 \rho \sigma_1 \sigma_2$$

Geometric
 $\rightarrow$ Orthogonality $\equiv$ Independence $\Leftarrow$ Prob.
 $a \cdot b = 0$ $\rho = 0$

$$\rho = 0 \Rightarrow \text{Var}[Y] = \sigma_1^2 + \sigma_2^2$$

$$\rho = 1 \Rightarrow \text{Var}[Y] = (\sigma_1 + \sigma_2)^2$$

$$\rho = -1 \Rightarrow \text{Var}[Y] = (\sigma_1 - \sigma_2)^2$$



Mean and Variance propagation laws

If $Y = a_1 X_1 + a_2 X_2 + c$, with a_i and c deterministic const.

$$E[Y] = a_1 \mu_1 + a_2 \mu_2 + c$$

$$\text{Var}[Y] =$$

Propagation: μ and σ

- Two structural engineers, Alice and Bob, independently measure the wind speed in m/s at the top of each tower.

Alice's reading $X \sim \mathcal{N}(\mu = 40, \sigma = 5)$ *Marginal of X*

Bob's reading $Y \sim \mathcal{N}(\mu = 42, \sigma = 6)$ *Marginal of Y*

Their readings are correlated with $\rho = 0.8$

- If Alice and Bob average their measurements, $W = (X + Y)/2$, what are $E[W]$ and $\text{Var}[W]$?
- What's the covariance of X and Y ?
- If design wind speed is 55 m/s, what's the prob. that the average exceeds 55 m/s?



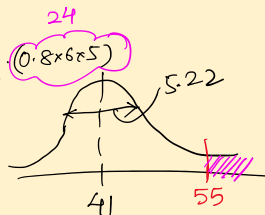
Propagation: mean and uncertainty

$$W = \underbrace{\frac{1}{2}}_{(a_1)} X + \underbrace{\frac{1}{2}}_{(a_2)} Y$$

$$\mu_W = \frac{1}{2} 42 + \frac{1}{2} 40 = 41$$

$$\sigma_W^2 = \left(\frac{1}{2}\right)^2 5^2 + \left(\frac{1}{2}\right)^2 6^2 + 2 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot (0.8 \times 6 \times 5)$$

$$\Rightarrow \sigma_W = 5.22$$



$$\begin{aligned} P(W \geq 55) &= 1 - P(W < 55) \\ &= 1 - \Phi\left[\frac{W - \mu}{\sigma}\right] = 1 - \Phi\left[\frac{55 - 41}{5.22}\right] \\ &= 1 - \Phi[2.68] = 1 - 0.9963 = 0.0037 \\ &= 0.37\% \end{aligned}$$

$$Y = a_1 X_1 + a_2 X_2$$

$$\mu_Y = E[Y] = a_1 \mu_1 + a_2 \mu_2$$

$$\sigma_Y^2 = E[(Y - \mu_Y)^2] = a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + 2a_1 a_2 \rho_{12} \sigma_1 \sigma_2$$

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Standard Normal Probabilities

Z-table

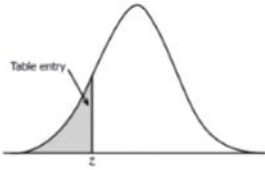


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Mean and Variance propagation laws

If $Y = a_1X_1 + a_2X_2 + \dots + a_nX_n + c$, with a_i & c det. consts.

$$E[Y] = a_1\mu_1 + a_2\mu_2 + \dots + a_n\mu_n + c$$

$$\begin{aligned}
 \text{Var}[Y] &= E[(Y - \mu_Y)^2] \\
 &= E[(a_1X_1 - a_1\mu_1) + (a_2X_2 - a_2\mu_2) + \dots + (a_nX_n - a_n\mu_n)]^2 \\
 &= a_1^2 E[(X_1 - \mu_1)^2] + \dots + a_n^2 E[(X_n - \mu_n)^2] \\
 &\quad + 2a_1a_2 \text{cov}(X_1, X_2) + 2a_1a_3 \text{cov}(X_1, X_3) + \dots + 2a_{n-1}a_n \text{cov}(X_{n-1}, X_n) \\
 &= \sum_{i=1}^n a_i^2 \sigma_i^2 + 2 \sum_{i < j} a_i a_j \rho_{ij} \sigma_i \sigma_j
 \end{aligned}$$

Mean and Variance propagation laws

- If $Y = g(X)$ is a nonlinear function of X . Find $E[Y]$ & $\text{Var}(Y)$.

$$E[Y] = E[g(X)]$$

linearize?

Taylor Series

$$g(x) = g(\mu_x) + \left. \frac{\partial g}{\partial x} \right|_{\mu_x} (x - \mu_x) + \frac{1}{2!} \left. \frac{\partial^2 g}{\partial x^2} \right|_{\mu_x} (x - \mu_x)^2 + \dots \text{H.O.T.}$$

2nd order approximation $E[Y]$.

$$E[g(X)] = g(\mu_x) + 0 + \frac{1}{2} \left. \frac{\partial^2 g}{\partial x^2} \right|_{\mu_x} \sigma_x^2$$

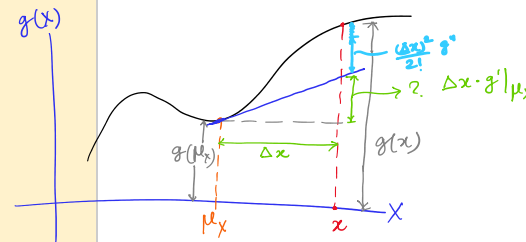
1st order approximation of $E[Y]$.

$$E[g(X)] = g(\mu_x)$$

1st order approximation of $\text{Var}[Y]$.

$$\text{Var}[g(X)] = \left(\left. \frac{\partial g}{\partial x} \right|_{\mu_x} \right)^2 \sigma_x^2$$

$$x \quad \frac{x^2}{2} \quad \frac{x^3}{3 \cdot 2} \quad \frac{x^4}{4 \cdot 3 \cdot 2}$$



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Mean and Variance propagation laws

- If $Y = g(X)$ is a nonlinear function of X . Find $E[Y]$ & $\text{Var}(Y)$.

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Example

- If $Y = \ln X$, what are expectation and variance of Y ? Use Taylor series approximations.

FO mean approx.

$$E[Y] \approx \ln \mu_x$$

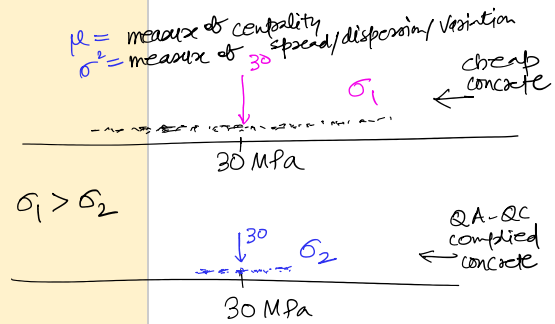
SO mean approx.

$$E[Y] \approx \ln \mu_x - \frac{1}{2} \frac{\sigma_x^2}{\mu_x^2} = \ln \mu_x - \frac{1}{2} \delta_x^2$$

FO variance approx.

$$\text{Var}(Y) \approx \delta_x^2$$

$$\text{C.O.V. } \delta_x = \frac{\sigma_x}{\mu_x}$$



$$E[Y] \cong g(\mu_x) + \frac{1}{2} \left(\frac{\partial^2 g}{\partial x^2} \right)_{\mu_x} \sigma_x^2 \quad \text{Second-order mean approximation}$$

$$\text{Var}[Y] = E[(Y - \mu_Y)^2] \cong \left(\left(\frac{\partial g}{\partial x} \right)_{\mu_x} \right)^2 \sigma_x^2 \quad \text{First-order var. approx.}$$

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Questions, comments,
or concerns?

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